

Scope Statement

Project Justification:

We are making this escape game because it is for our final project for grade 12 computer science. The in person experience of an escape room has attracted many people over the past decade, it is a fun and alluring challenge. It's become so popular that computer games have been created to simulate the experience. We were inspired to make an escape style game by a similarly themed game on cool math games. We found the idea fun and quickly realized that it can be adapted to different situations that we encounter. This game idea gives us the flexibility that we need to overcome obstacles such as over-estimating time constraints, or the complexity of certain ideas. It is a collection of smaller games that we can control the intricacy of based on changing time restraints or goals. Instead of worrying if one big game is functioning, our game is split into essentially smaller games that can be easily removed or changed in a time crunch.

Project's Products:

The player is stuck in a room and their goal is to escape. They do so by walking around the room and inspecting objects. Objects will lead to mini games such as puzzles and memory games. The completion of mini games will award the player with new clues or keys to logically discover the way out. The minigames themselves will all be unique to challenge the player in different ways. For example, we would have a memory game to help with the players focus and concentration while other games might have timers as they do certain tasks to test the players overall skills. We would also have mazes and riddles scattered throughout the game to challenge the players critical thinking skills.

Project Deliverables:

- Functioning game
- Developed storyline
- Instructions
- Main menu
- Documentation containing solutions and hints (cheat sheet)
- Reflection summary

Project Objectives:

- Getting 100% on the project
- At least 3 completed mini games
- at least 3 objects to interact with
- a successful way to escape
- one moveable character to "walk" around the room (can move up down left right)
- A game that challenges the player and takes at least 10 minutes to complete
- All the objects and minigames logically connect
- Looks polished (includes non interactive objects to fill the space)

WBS

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1.1 Project Charter

The need the project addresses:

- Our “client” requested a 2D graphical game in Java

The product description:

- Escape room game. Find clues and complete puzzles to find a way to escape the room

The Project Manager Identified/Assigned:

- Luka

Constraints (A list of factors that will limit the project management team's options i.e. Time, Cost, Technology, Schedules etc):

- We would have to find a way to make puzzles without making them too easy or too hard (where to place items, minigames, types of puzzles)
- Graphics
- Prioritizing this over other classes (should we, when should we)
- Everyone's schedules compared to how much work they need to get done
- Must be done in class or at least on a desktop

Assumptions (A list assumptions you have made about the project i.e. Knowledge base, time, abilities, technology etc):

- The minigames will take the most of the time and the actual room will be easy to make since it requires a number of images and only requires a mouse. (must learn graphics before designing rooms)
- Our graphics knowledge will be limited
- Our combined knowledge of coding is also limited
- Renee
 - Courses: Chem, Comp Sci, Adv Functions
 - Job: 9 hours a week, Busy on Tuesdays, Fridays, Sundays
 - Naps take 2 hours on average (it also takes me half an hour to recover from my naps they take a lot out of me)
- The features might be challenging to code and we might have to scrap some of them
- It will be difficult to figure out the logic of tying the mini games together and the intended route of the room (planning)
- It will probably take a lot of planning to ensure that the user doesn't go off the route and skip ahead in puzzles (can plan puzzles outside class)

A list of Key Stakeholders:

- Ms Karaisinska
- Luka
- Renee
- Andrew

1.2 Team Norms

Team member interactions

- Team members are all treated with mutual respect
- Don't be reluctant to offer feedback or pitch new ideas to the group
- Big changes or conversations are done in person, not in the group chat
- Check the group chat regularly for updates and questions

- Any changes that you want to make or need to be made are discussed with the group
- Always make sure to include everyone in the group
- If there is something that could affect the progress of the group don't be reluctant to let everyone know (especially if it's your schedule or personal issues)
- Group members don't hesitate to help out their other members
- Let the rest of the group know that you will be absent as soon as possible

Expectations

- Everyone is keeping up to date with their portion of the work. If they fall behind they do their best to catch up or reach out to other team members if they need help
- Make sure to get at least a few hours of work in each week
- Everyone is meeting the groups set deadlines for their work
 - Deadlines and other important dates will be further discussed as we move through the course
- Everyone is keeping the rest of the group updated with their progress and any problems that may arise
- No playing games during class time

Team problem solving, conflict resolution and decision making

- Tattle tailing to Ms. K is always an option
- Admitting when you're wrong shows more strength
- If people can't agree on something, come up with a new solution or compromise.
- If a compromise can't be made, either consult with Ms. K or change the idea entirely

1.4 Risk Management

1) Team Issues

Category: Productivity issues

Description: Sometimes someone might get annoyed by someone else potentially reducing the productivity of the team until the issue is resolved.

Risk probability: 0.9

Risk impact: 0.05

Risk score: 0.045

Response plan: This risk wouldn't impact the project too much as we would be in class and should be working anyways. We should always refer back to the team norms.

2) Sickness

Category: Productivity issues

Description: Someone gets sick for an extended period of time and it makes it difficult for that person to finish their parts of the project.

Risk probability: 0.7

Risk impact: 0.4

Risk score: 0.28

Response plan: Keep communication over messages so everyone is always up to date, sick person can continue to work from home if they are able. If unable to get work done, the work will be re-allocated to the other members.

3) Group members dropping the class

Category: Employee turnover

Description: Someone might drop the class as the semester progresses.

Risk probability: 0.1

Risk impact: 0.4

Risk score: 0.04

Response plan: Attempt to divy up the work again. If there is too much work to split between the remaining members try to simplify the project and redefine the scope.

4) Group members run into technical difficulties with their personal computers

Category: Procedural risk

Description: Sometimes group members can run into technical difficulties at home, such as losing wifi, computer broken, computer doesn't function properly or efficiently. These issues could last from a couple minutes to couple days or even weeks if not dealt with. Risk impact would vary depending on what the issue is specifically and how long it lasts for.

Risk probability: 0.5

Risk impact: 0.2

Risk score: 0.1

Response plan: Keep communication over messages so everyone is always up to date. Person tries their best to use class time to get it done or come in during lunch or spare if needed. If unable to get work done, the work will be re-allocated to the other members.

5) Underestimate how much time we will need to complete certain tasks

Category: Estimating and scheduling

Description: Certain tasks could take longer than expected requiring us to shift our workflow around it. It can affect linear processes that require previous steps.

Risk probability: 0.7

Risk impact: 0.4

Risk score: 0.28

Response plan: As previously mentioned we can change our workflow to focus on what we need to have and if we need to simplify some aspects of our games. We can also go over everything that needs to be done and reschedule and redistribute the work to group members.

6) Adding unnecessary features to the game

Category: Gold plating

Description: Sometimes we can get carried away with spending time on parts of the game that could be unnecessary. Especially if it takes way more time than anticipated.

Risk probability: 0.5

Risk impact: 0.4

Risk score: 0.2

Response plan: If something is taking much longer than expected, revisit the importance of the feature and if we can scale it back or cut it completely

7) Spending too much time on design and not leaving enough time to code the game

Category: Gold plating

Description: Especially for the game we have chosen, there are a lot of non-coding decisions that need to be made. Such as the storyline, or the design of the objects. This could affect the amount of time we have to actually code.

Risk probability: 0.5

Risk impact: 0.4

Risk score: 0.2

Response plan: If it is taking too long to decide, start on the areas that are certain and work from there.

8) Off task

Category: Productivity issues

Description: Group member gets off task either at home or in school

Risk probability: 0.9

Risk impact: 0.4

Risk score: 0.36

Response plan: Try to figure out if they are struggling with something. Generally, procrastination is a result of lack of understanding so if the missing piece of the puzzle is learnt, the group member might be more motivated to keep working.

9) Combining Code Doesn't Work

Category: Technical Risk

Description: Since we plan on using Google Drive to store files, when uploading changes, we need to replace the entire file. This can cause issues if group members happen to be working on the same file.

Risk probability: 0.5

Risk impact: 0.6

Risk score: 0.3

Response Plan: We should make extra backups for files and upload them to a new location. That way, the old files will always remain and we can figure out other ways of combining the versions such as copying the changes instead of the whole document.

10) Spending Too Little Time on Design and Not Enough on Code

Category: Compromising on Designs

Description: If too much time is spent on creating elaborate designs, we might not have enough time to code our vision.

Risk probability: 0.3

Risk impact: 0.8

Risk score: 0.24

Response Plan: Set a hard limit on when design has to be finished. That way, we are guaranteed a large amount of time for coding. Alternatively, we can have 2 people start what we already planned while the last person finishes up the design work.

2.1 Use Cases

Main Game Play

Player clicks play Game

- A) Game throws Player into the main room

Player walks through door

- A) Player walks through door
- B) Room that is shown changes

At A, if door is locked, player will not be able to walk through door

Player unlocks door

- A) When a player moves close enough, they are prompted with a message to unlock the door. Player can only unlock door when message appears
- B) Door changes to open state

At A, if the user is not close enough to the door or does not have the right key, a message will not appear.

Player moves

- A) Player used the WASD keys to move around the room

At A, if user tries to move past the borders of the room, player will stop moving

Player uses item in inventory

- A) When a player moves close enough, they are prompted with a message to be able to use the item. They are only able to use item once message appears

At A, if item can't be used at the right object, no message will appear

Collect item

- A) When a player moves close enough, they are prompted with a message to be able to collect an item. They are only able to collect an item once the message appears

Inspect environment

- A) When a player moves close enough to a part of the room they can inspect, they are prompted with a message to be able to inspect the environment. They are only able to inspect environment once the message appears

Place item

- A) When player inspects stand, they will be prompted with a message to be able to place item in the stand

At A, if they don't have the right items in their inventory, no message will appear

Memory Game

Player interacts with game tile

- A) When player is on a tile, they are prompted with a message to light up the tile that they are standing on

At A, if the tile is not correct, reset the memory game (all tiles flash twice)

Slide Box Puzzle Game

Player slides boxes

- A) Player selects box
- B) Player uses arrow keys to move the box either left, right, up or down

At A, if there is another box in the way, box will not be able to move in that direction

At A, if player tries to move box outside of the border, box will not be able to move in that direction

At A, once box moves, no box will be selected

Player gets red box out of area

- A) Player is given second key

Reset button

- A) Player clicks on button and game restores back to its original state

Spot the Difference

Player selects a difference

- A) Player clicks a spot to circle the difference

At A, if player clicks on an area that is already circled do not circle again

At A, if player clicks last difference, give key number one and use case ends

2.3.1 Character Design



The user's character and any interactive characters throughout the game will follow this design style. We will give the player a choice of three characters so that they can play with the preferred one.

2.3.1 Storyline Design / Gameplay Loop

Our game will follow the premise of letting the character control a thief. They got into a museum and are looking for a jewel. However, the jewel as well as their exit are locked behind a vault requiring 3 keys. To beat the game, the players must unlock the vault using three keys which are obtained through minigames.

Players will see the following story if they want to: "You are in an empty museum searching for riches. Unfortunately, the only way out is through a vault door. On the bright side, that was your goal in the first place. Sadly, it requires 3 keys. Look for clues and obtain items to unlock the door.". Additionally, the player is shown the controls when they hit play: "Use WASD to move around. Press E to interact with objects. Use the mouse to select / pickup items.".

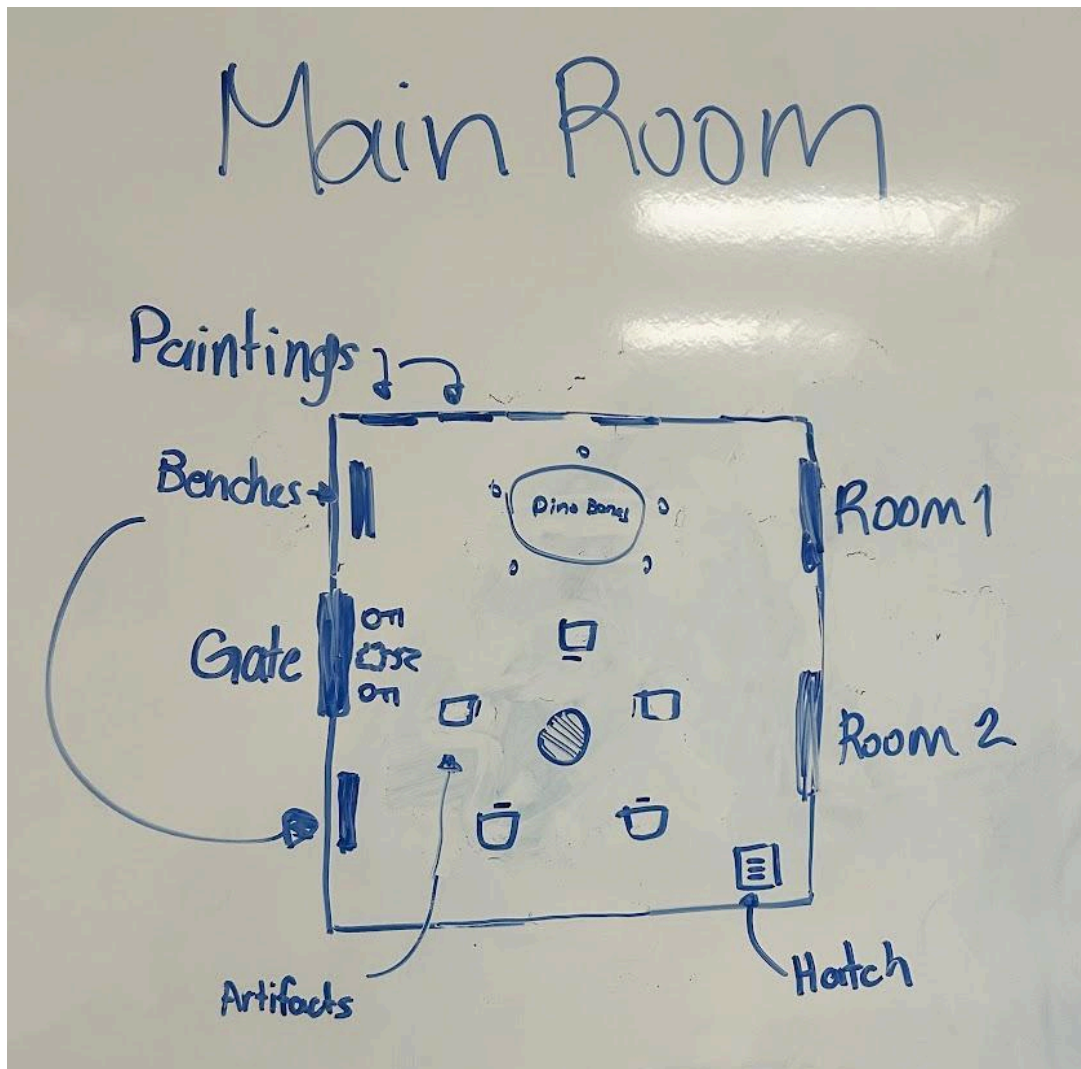
Afterwards, their character is dropped into the first of 3 rooms which we call the main room. Inside the main room, there are 3 doors, the vault door, the door to a locker room and a basement. All three doors are locked. We designed the game to have each room give the player a key for the vault as well as a way of progressing to the next room. In the first room, interacting with a painting opens a "spot the difference" game. Beating the game will give the player a vault key. Additionally, the pedestals arranged in a pentagon shape is also a light up game in which the player lights up the pedestals to match the dinosaur bones in the sandbox above. This gives the player another key, this time to unlock the locker room door.

In the locker room, there is a bench with a stone tablet on top as well as lockers. The lockers are openable with one containing a slidey box puzzle which also awards a vault key when

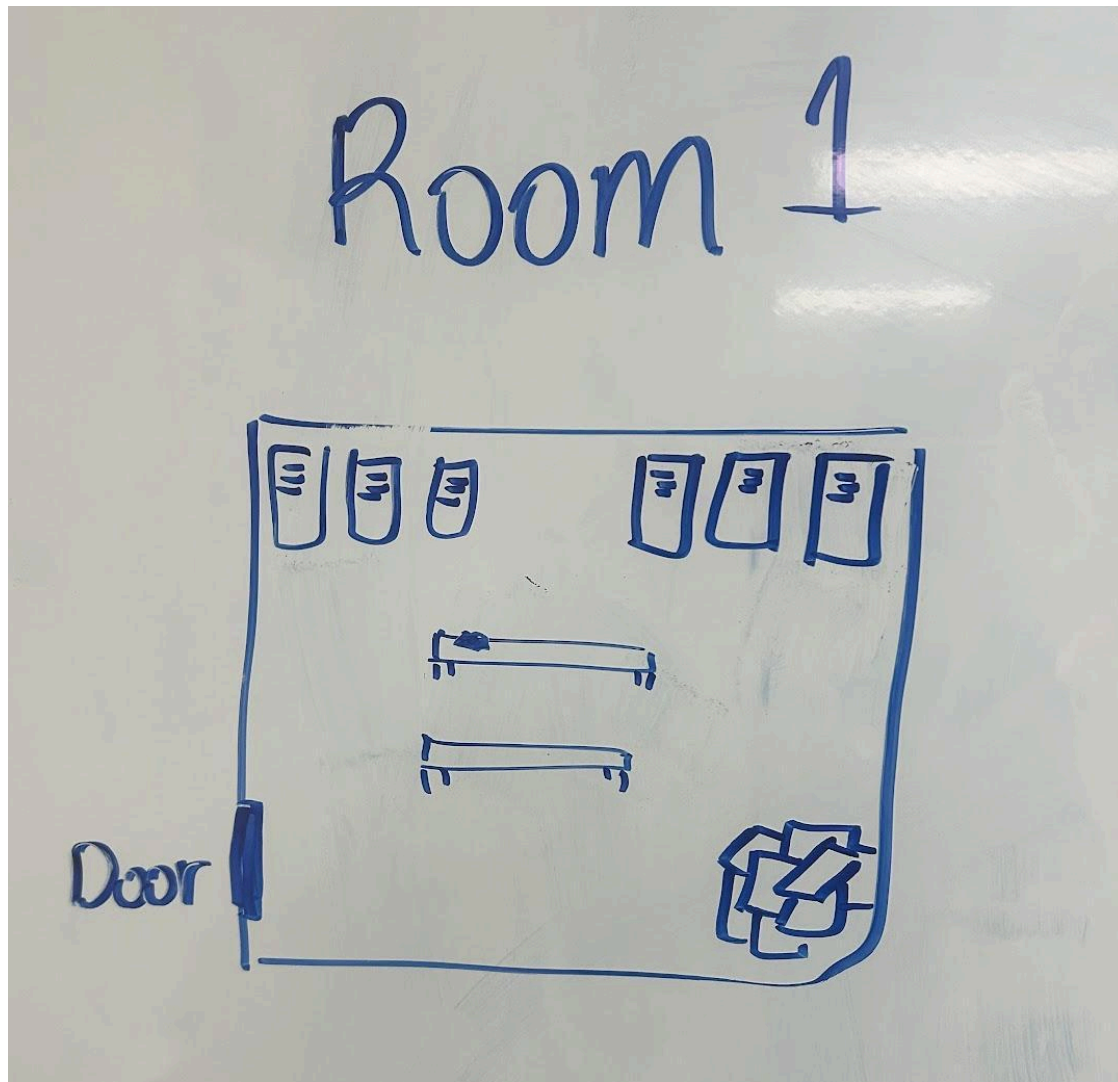
beaten. The other lockers will contain a shovel and another tablet piece. These tablet pieces can be put into a stone stand in the main room to eventually form a decoder to unlock the last room. However, 3 of them are needed. The 3rd one is obtained by using the shovel and interacting with the sandbox which digs up the necessary tablet.

This means only the last room is left. The last room has a door secured by a combination lock. The combination is obtained by using the stone tablet decoder. Once in the room, there is a button as well as what looks like a dance pad. By interacting with the button, the player starts up a memory game in which they have to remember which squares on the 4 by 4 grid light up. After pressing the button, they are unable to leave the room unless they beat the game and obtain the last vault key or lose. After obtaining the final key, the player can complete the game by putting the keys into the vault door.

2.3.3 Room Designs



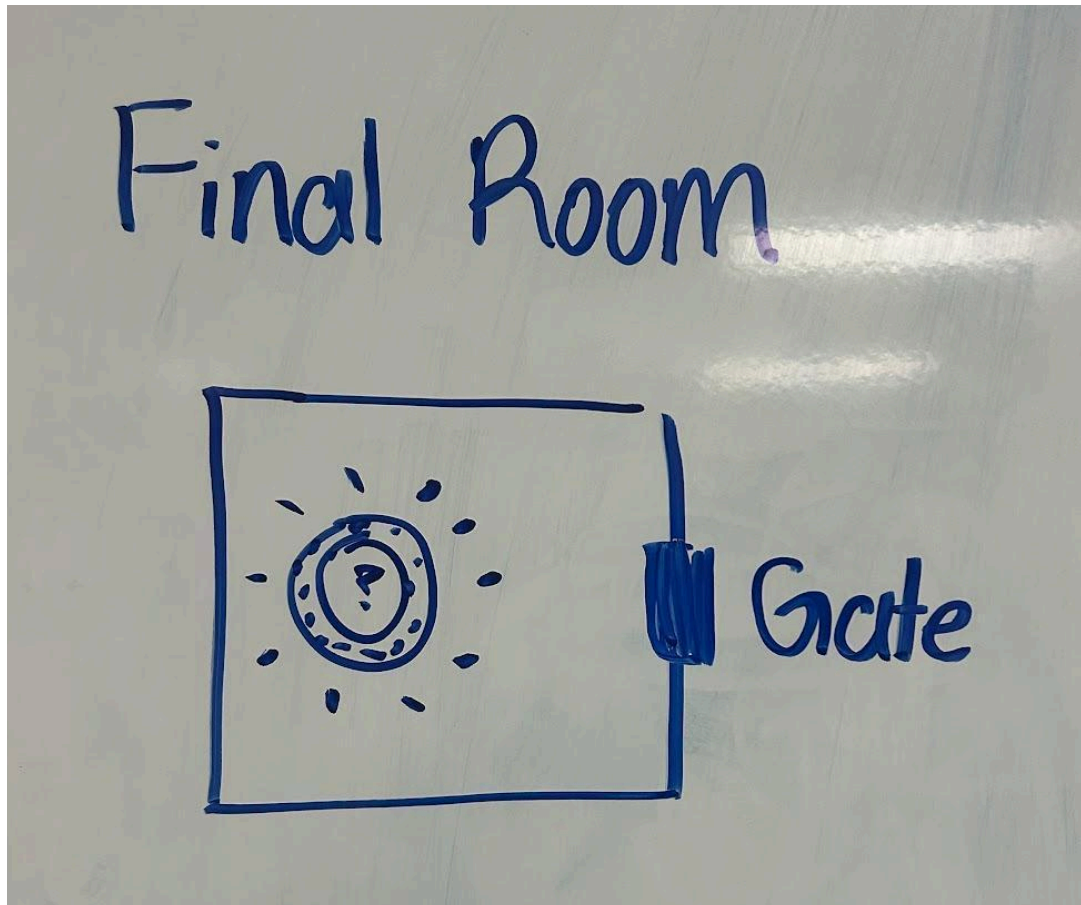
The first room we have is the main room. Inside we have the main gate on the left wall, which leads to the final room the user is trying to break into. On the right wall, there are two locked doors. One of them leads to room 1, and the other leads to room 2, both of which will provide the user with clues and tools to help them get through the main gate. There is a possibility of a 3rd room if we have time, that works the same as room 1 and 2 but instead of it being a door the user has to unlock, it is a hatch that leads to the bottom floor. Around the room are a bunch of objects and design elements to make it look like a museum while also providing us, the developers, places to put clues and other puzzles. For example, we have benches on the left side of the room, 4 paintings on the back wall, a sand pit with 5 lights surrounding it in the top half of the room and 5 display cases in the bottom half of the room. The circle that's in the middle of the display cases in our diagram represents the object that will come out of the floor once the user completes the puzzle. The hatch ended up being scrapped as we moved along in our design due to the fact we already had more than enough minigames.



Room 1 is the first room that the user will be able to obtain a key that helps unlock the main gate. In this room we have a pile of boxes in the bottom right corner, two benches in the middle of the room, 3 lockers in the top left corner and 3 lockers in the top right corner. This inspiration for this room is supposed to be the security guard room.



Room 2 is the next room that gives the player the opportunity to obtain a key to open the main gate. In this room, there isn't anything except for a 4x4 grid on the floor. This grid is supposed to be a memory and once the user beats that game, the 2nd key will come up from the floor.

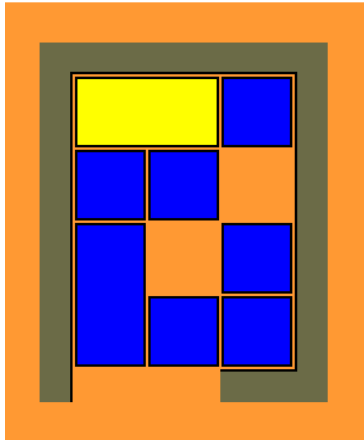


A fancy shmancy room that makes the user feel special for finding it. In the middle there is the final object on a pedestal with a spotlight. In the end, we decided on not showing the fancy room and just having an ending screen instead as the room didn't add too much.

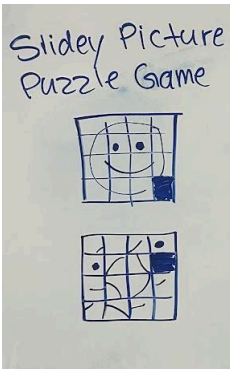
2.3.4 Puzzles and Mini Games Design



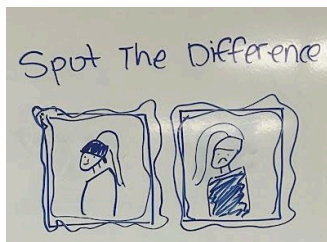
All the boxes can slide left, right, up, and down by the user. The goal is to slide the coloured box to the exit by moving the other ones out of the way.



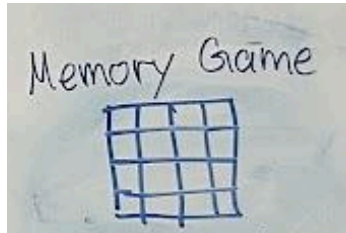
This is the layout we will use for the slidey box game.



Has similar mechanics to slidey box game, except now there is an image that is scrambled. The goal is to slide the square such that they are arranged to complete the image. There is one empty square that allows one box to be slid at a time. This ended up being scrapped as we already had 3 games that gave out a total of 3 keys.

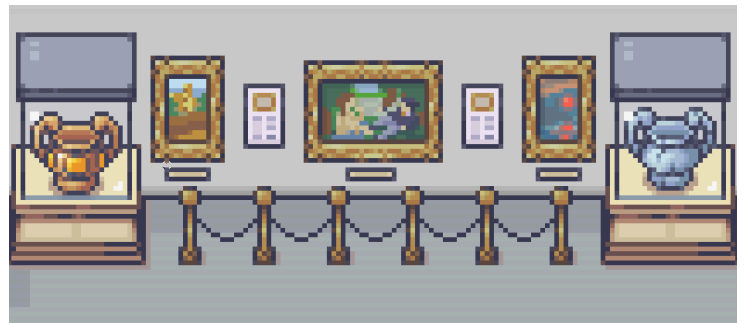


In the main room there will be various paintings on the far wall. Two paintings will look similar but will have variance. The player needs to identify this and zoom in on them, a text will appear that hints to the user that they should spot the difference. The user can click on certain parts of the pictures and it will circle the differences on each one. Once all of the differences have been found and circled the door to room 1 will open.



Squares will light up in a certain sequence, the user has to memorize said sequence and click/stand on them. The sequence will get longer after each completed "level".

2.3.5 Object Design



The pictures above are meant to show the type of style that we want to include in the game and shows a little bit what the museum would look like.